Spring 2026 Topic Reviews for Math 54

Faculty:	Profs. Arun Sharma and Vera Serganova
Facilitator:	Jason Ferguson, jason.ferguson@berkeley.edu
Time and Location:	Tu 3:30-5 on Zoom and in 201A Student Center, Th 3:30-5 on Zoom and 6 Evans
Jason's Office Hours:	As of 3/16, M 12-1, TuTh 2-3 See here for the most up-to-date Drop-In Schedule

Worksheet for 3:30-5pm Tuesday, May 5, 2026's Math 54 Topic Review

Overall Topic: Fourier sine and cosine series, performing computations accurately and efficiently (B10.4, some 5.1, 5.5, 6.3, 6.4, 6.7, B10.3)

Objectives: By engaging actively with this session, participants will make progress towards being able to:

10.4

1. Given an appropriate function defined on $[0, L]$, compute its Fourier cosine or sine series
2. Given a function defined on $[0, L]$, identify its odd extension to $[-L, L]$ and even extension to $[-L, L]$ and use both to identify the functions that its Fourier sine and cosine series converge to.
3. Accurately and efficiently perform the following two key calculations from Math 54:
   (a) Find eigenvectors of a 3×3 or larger matrix once the eigenvalue is known, including complex eigenvalues.
   (b) Gram-Schmidt and find orthogonal projections, for both dot products and for general inner products
4. Using guidelines to identify for themselves what additional topics would benefit them to study.

Contents

Classifying types of questions on this worksheet

I thought it would be good for your studying to divide the problems on this worksheet into the following three types:

- **R**: Recall: You need to use only one main concept or method to solve the problem, in the way it's presented in the textbook or lecture notes.
- **A**: Analyze: You need either one main concept or method to solve the problem but in a way that's different from how it's typically presented, or you need to use a group of related ideas to solve the problem.
- **S**: Synthesize: You need to combine multiple concepts that weren't originally shown to be related to solve this problem.

3:45

(10.3-10.4) Review of Fourier series,Fourier sine series, and Fourier cosine series from 4/30's Review

Definition: (Book) Let $L > 0$ be a constant. Let f be a piecewise-continuous function on the interval $[-L, L]$. The **Fourier Series** of f is the infinite (trigonometric) series

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

where:

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x) \cancel{\cos\left(\frac{0\pi x}{L}\right)}^{1} dx = \frac{1}{L}\int_{-L}^{L} f(x)\,dx$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x) \cos\left(\frac{n\pi x}{L}\right) dx \text{ for } n = 0, 1, 2, \ldots \quad \text{and} \quad b_n = \frac{1}{L}\int_{-L}^{L} f(x) \sin\left(\frac{n\pi x}{L}\right) dx \text{ for } n = 1, 2, \ldots.$$

Abbreviation: "The Fourier series of $f(x)$ is $\frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right]$" is shortened to:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right].$$

Definition: Let $L > 0$ be a constant. Let f be a piecewise-continuous function on the interval $[0, L]$. The **Fourier Cosine Series** of f is the infinite (trigonometric) series

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) \qquad \text{where} \qquad a_n = \frac{2}{L}\int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \text{ for } n = 0, 1, 2, \ldots.$$

The **Fourier Sine Series** of f is the infinite (trigonometric) series

$$\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) \qquad \text{where} \qquad b_n = \frac{2}{L}\int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \text{ for } n = 1, 2, \ldots.$$

1. **A** In your own words, in one sentence:
   (a) (Review from 4/30's Review) What does it mean to represent a function on $[-L, L]$ as a Fourier series?
   (b) What does it mean to represent a function on $[0, L]$ as a Fourier cosine series?
   (c) What does it mean to represent a function on $[0, L]$ as a Fourier sine series?

(a) Orthogonal projection of f as an infinite sum of sines & cosines. (and constant).

OR

Best approximation to f as an infinite sum of sines & cosines & constant

(b) Best approximation to f as an infinite sum ~~of sines~~ & cosines & constant

(c) Best approximation to f as an infinite sum of sines ~~& cosines & constant~~

(10.4) Computing Fourier Sine and Cosine Series with Tanya

Reminders: (Book) Let $L > 0$ be a constant. Let f be a piecewise-continuous function on the interval $[0, L]$.

Fourier Cosine Series on $[0, L]$: $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$ where $a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)dx$ for $n = 0, 1, 2, \ldots$.

Fourier Sine Series on $[0, L]$: $\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$ where $b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)dx$ for $n = 1, 2, \ldots$.

$f(x) = 1$

2. **R** Let $f(x)$ be the constant function 1. For each part, try to do as little work as you need to find an answer.

   (a) (Review from 4/30's Review) Find the Fourier series of f on $[-\pi, \pi]$.

   (b) Find the Fourier cosine series of f on $[0, \pi]$.

   (c) Find the Fourier sine series of f on $[0, \pi]$.

$L = \pi$

a) $f(x) = 1 \sim 1 + \sum_{n=1}^{\infty} 0 \cdot \cos(nx) + 0 \cdot \sin(nx)$

(Longer: Show $\frac{a_0}{2} = 1$)

$a_n = 0$ for all $n \geq 1$

b) $f(x) = 1 \sim 1 + \sum_{n=1}^{\infty} 0 \cdot \cos(nx)$

c) Will $b_n = 0$? No! Can't use $\int_{-L}^{L}$ odd $= 0$

Also: Doesn't make sense $f(x) = 1 \sim 0$

See the following graph to see both $f(x)$ on $[0, \pi]$ and partial sums of the Fourier sine series of f:
https://www.desmos.com/calculator/xck8ge29yi

Solve for b_n ü

The **Fourier Sine Series** of f is the infinite (trigonometric) series

$$\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) \qquad \text{where} \qquad b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)dx \text{ for } n = 1, 2, \ldots .$$

$$b_n = \frac{2}{\pi}\int_0^{\pi} 1 \cdot \sin(nx)\, dx$$

$$= \frac{2}{\pi}\left[-\frac{\cos(nx)}{n}\right]_0^{\pi}$$

$$= \frac{2}{\pi} \cdot -\frac{1}{n}\left[\cos(\pi n) - 1\right]$$

$$b_n = -\frac{2}{\pi n}(\underbrace{\cos(\pi n)}_{(-1)^n} - 1)$$

Extra: Notice

$$b_n = \frac{2}{\pi n}\left(1 + (-1)^{n+1}\right)$$

$$b_n = \begin{cases} 0 & , \ n \text{ even} \\ \frac{4}{\pi n} & , \ n \text{ odd} \end{cases}$$

Extra practice for finding Fourier cosine and sine series

3. Find the Fourier cosine and sine series for the given function $f(x)$ on the given interval.

   (a) **R/S** $f(x) = 2 - 5\cos(3x)$ on $[0, \pi]$
   (b) **S/R** $f(x) = 3\sin(2\pi x) - 4\sin(6\pi x)$ on $[0, 2]$
   (c) **R** $f(x) = x$ on $[0, L]$
   (d) **R** $f(x) = x + 1$ on $[0, \pi]$
   (e) **A** $f(x) = x^2$ on $[0, L]$
   (f) **A** $f(x) = e^x$ on $[0, L]$
   (g) **A** $f(x) = x(L - x)$ on $[0, L]$

(a) $\boxed{f \text{ is its own Fourier cosine series}}$ $\boxed{\text{i.e. } \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) \text{ where } a_0 = 4,\ a_3 = -5,\ a_n = 0 \text{ for all other } n.}$

$$\boxed{\text{Sine series: } \sum_{n=1}^{\infty} b_n \sin(nx) \text{ where } b_n = \begin{cases} \frac{8}{n\pi} & \text{if } n \text{ is odd.} \\ -\frac{20n}{\pi(n^2-9)} & \text{if } n \text{ is even} \end{cases}}$$

$$\text{Less simplified } b_n: = \begin{cases} \frac{6n^2+36+(-1)^n(14n^2-36)}{\pi n(n^2-9)} & \text{if } n \text{ any positive integer \underline{except} } 3 \\ \frac{8}{3\pi} & \text{if } n=3 \end{cases}$$

(b) $\boxed{f \text{ is its own Fourier sine series}}$

$$\boxed{\text{Cosine series: } \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{2}\right) \text{ where } a_n = \begin{cases} -\frac{48}{\pi(n^2-16)} + \frac{192}{\pi(n^2-144)} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even.} \end{cases}}$$

$$\text{Less simplified } a_n = \begin{cases} \frac{24((-1)^n-1)}{\pi(n^2-16)} - \frac{96((-1)^n-1)}{\pi(n^2-144)} & \text{for all } n \geq 0 \text{ except } n = 4 \text{ or } 12 \\ 0 & \text{for } n = 4 \text{ \& } 12. \end{cases}$$

(e) – (g) on next page

4. **A** To solve parts (a) and (b) of Problem 3 on the interval $[0, L]$ for general $L > 0$, you'd need to divide into cases based on what L is. Why?

5. **S** Solve part (a) of Problem 3 on the interval $[0, L]$ for general $L > 0$. [You're welcome to try (b), as well, but it will be even messier than (a) because there are two different trig terms.]

Extra practice for finding Fourier cosine and sine series

3. Find the Fourier cosine and sine series for the given function $f(x)$ on the given interval.

   (a) **R/S** $f(x) = 2 - 5\cos(3x)$ on $[0, \pi]$
   (b) **S/R** $f(x) = 3\sin(2\pi x) - 4\sin(6\pi x)$ on $[0, 2]$
   (c) **R** $f(x) = x$ on $[0, L]$
   (d) **R** $f(x) = x + 1$ on $[0, \pi]$
   (e) **A** $f(x) = x^2$ on $[0, L]$
   (f) **A** $f(x) = e^x$ on $[0, L]$
   (g) **A** $f(x) = x(L - x)$ on $[0, L]$

(c) $\boxed{\frac{L}{2} + \sum_{n=1}^{\infty} \frac{2L((-1)^n - 1)}{\pi^2 n^2} \cos\left(\frac{n\pi x}{L}\right)}$ $\boxed{\sum_{n=1}^{\infty} \frac{2L}{\pi n}(-1)^{n+1} \sin\left(\frac{n\pi x}{L}\right)}$

(d) (Answer to 3c with $L = \pi$) + (Answers to 2b & 2c)

$\boxed{\left(1 + \frac{\pi}{2}\right) + \sum_{n=1}^{\infty} \frac{2((-1)^n - 1)}{\pi n^2} \cos(nx)}$ $\boxed{\sum_{n=1}^{\infty} \left[\frac{2}{n\pi}\left[1 - (-1)^n\right] + \frac{2}{n}(-1)^{n+1}\right] \sin(nx)}$

(e) $\boxed{\frac{L^2}{3} + \sum_{n=1}^{\infty} \frac{4L^2}{\pi^2 n^2}(-1)^n \cos\left(\frac{n\pi x}{L}\right)}$ $\boxed{\sum_{n=1}^{\infty} \frac{2L^2\left[(2 - \pi^2 n^2)(-1)^n - 2\right]}{\pi^3 n^3} \sin\left(\frac{n\pi x}{L}\right)}$

(f) $\boxed{\frac{e^L - 1}{L} + \sum_{n=1}^{\infty} \frac{2L\left[(-1)^n e^L - 1\right]}{L^2 + \pi^2 n^2} \cos\left(\frac{n\pi x}{L}\right)}$ $\boxed{\sum_{n=1}^{\infty} \frac{2n\pi\left[1 - (-1)^n e^L\right]}{L^2 + n^2\pi^2} \sin\left(\frac{n\pi x}{L}\right)}$

(g) (Answer to (c))$\cdot L$ − (Answer to (e))

$\boxed{\frac{L^2}{6} + \sum_{n=1}^{\infty} \left(-\frac{2L^2((-1)^n + 1)}{\pi^2 n^2} \cos\left(\frac{n\pi x}{L}\right)\right)}$ $\boxed{\sum_{n=1}^{\infty} \left(-\frac{4L^2((-1)^n - 1)}{\pi^3 n^3} \sin\left(\frac{n\pi x}{L}\right)\right)}$

4. **A** To solve parts (a) and (b) of Problem 3 on the interval $[0, L]$ for general $L > 0$, you'd need to divide into cases based on what L is. Why?

In (a) If L is π, or 3π, or any number for which 3 is an integer times $\frac{L}{\pi}$, $f(x)$ is its own Fourier cosine series, but not for different L.

5. **S** Solve part (a) of Problem 3 on the interval $[0, L]$ for general $L > 0$. [You're welcome to try (b), as well, but it will be even messier than (a) because there are two different trig terms.]

Next page.

$f(x) = 2 - 5\cos(3x)$ on $[0, L]$

Cosine series:

If $L = \frac{N\pi}{3}$ for some positive integer N, then f is its own cosine series.

If not, $\left(2 - \frac{5}{3L}\sin(3L)\right) + \sum_{n=1}^{\infty} \frac{30(-1)^n L\sin(3L)}{n^2\pi^2 - 9L^2} \cos\frac{n\pi x}{L}$

$$\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) \quad \text{where } b_n = \begin{cases} \frac{4}{n\pi}(1-(-1)^n) + \frac{10n\pi\left[(-1)^n\cos(3L) - 1\right]}{n^2\pi^2 - 9L^2} & \text{if } n \neq \frac{3L}{\pi} \\ \frac{4}{n\pi}(1-(-1)^n) & \text{if } n = \frac{3L}{\pi} \text{ (not possible unless } L = \frac{N\pi}{3} \text{ for some } N\text{)} \end{cases}$$

4:07 **(10.4) Computing Fourier, Fourier cosine, and Fourier sine series for piecewise functions**

6. **A** Let $f(x)$ be the following function on $[0, 4]$: $\leftarrow L = 4$

$$f(x) = \begin{cases} x & \text{if } 0 \le x < 2 \\ 1 & \text{if } 2 \le x < 4 \end{cases}$$

Write down the integrals you'd need to compute to find b_n. (If you need the practice, evaluate the integrals and use them to find the Fourier sine series of f on $[0, 4]$. If you need more practice, also find the Fourier cosine series of f on $[0, 4]$.)

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

$$= \frac{2}{4}\int_0^4 f(x)\sin\left(\frac{n\pi x}{4}\right)dx$$

$$= \frac{2}{4}\left[\int_0^2 f(x)\sin\left(\frac{n\pi x}{4}\right)dx + \int_2^4 f(x)\sin\left(\frac{n\pi x}{4}\right)dx\right]$$

$$= \boxed{\frac{1}{2}\left[\int_0^2 x\sin\left(\frac{n\pi x}{4}\right)dx + \int_2^4 1\cdot\sin\left(\frac{n\pi x}{4}\right)dx\right]}$$

Answer: Sine series: $\displaystyle\sum_{n=1}^{\infty}\left[\overbrace{\frac{8}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right) - \frac{2}{n\pi}\cos\left(\frac{n\pi}{2}\right) - \frac{2}{n\pi}\cos(n\pi)}^{b_n}\right]\sin\left(\frac{n\pi x}{4}\right)$.

Cosine series: $\displaystyle 1 + \sum_{n=1}^{\infty}\left[\frac{8}{n^2\pi^2}\left(\cos\left(\frac{n\pi}{2}\right) - 1\right) + \frac{2}{n\pi}\sin\left(\frac{n\pi}{2}\right)\right]\cos\left(\frac{n\pi x}{4}\right)$.

The following graph shows $f(x)$ on $[0, 4]$ (green) and partial sums of the Fourier sine (purple) and cosine (orange) series of f: https://www.desmos.com/calculator/micmll86ia

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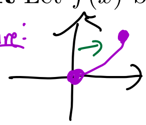
(10.4) Periodic, even, and odd extensions

7. **R** Let $f(x)$ be the function x^2 but with domain $[0, 1]$. Is f an even function?

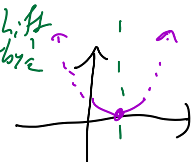
Picture:



$\boxed{\text{No.}}$

Algebra $f(1)=1$ $f(-1)=$ not defined, so $f(1)\neq f(-1)$.

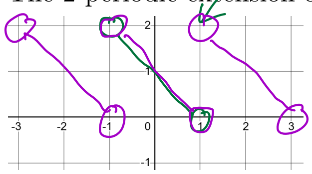
If you shift right by 2



8. **R/A** Let $f(x)$ be the function $1-x$ with domain $(-1, 1)$. Let $g(x)$ be the function $1-x$ with domain $(0, 1)$. Find graphs of each of the following functions for $-3 \le x \le 3$. (For extra practice if you need it, you can also find their formulas for $-3 \le x \le 3$.)

(a) The 2-periodic extension of f.

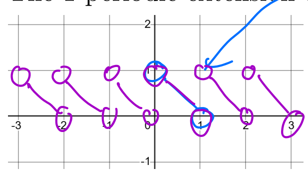
f in green



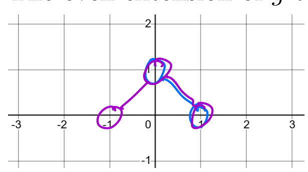
Answers in purple.

(b) The 1-periodic extension of g.

g in blue



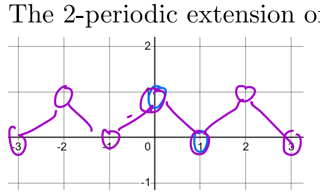
(c) The even extension of g to $(-1, 0) \cup (0, 1)$.



Also $g(|x|)$

$$g_e(x) = \begin{cases} g(-x) & \text{if } -L<x<0 \\ g(x) & \text{if } 0<x<L \end{cases} = g(|x|)$$

(d) The 2-periodic extension of the even extension of g to $(-1, 0) \cup (0, 1)$.



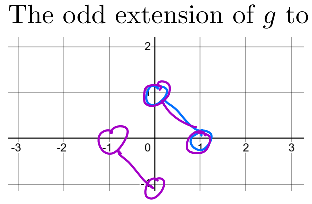
Also $g(|x|)$

$$g_o(x) = \begin{cases} -g(-x) & \text{if } -L<x<0 \\ g(x) & \text{if } 0<x<L \end{cases}$$

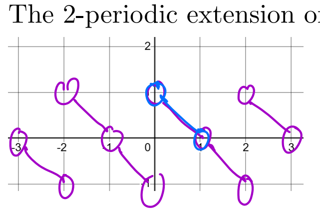
$$g_o(\underbrace{-\tfrac{1}{2}}_{x}) = -g\left(\tfrac{1}{2}\right) = -g\left(-(\underbrace{-\tfrac{1}{2}}_{x})\right)$$

$$g_o(x) = -g(-x) \quad \text{for } x<0$$

(e) The odd extension of g to $(-1, 0) \cup (0, 1)$.



(f) The 2-periodic extension of the odd extension of g to $(-1, 0) \cup (0, 1)$.



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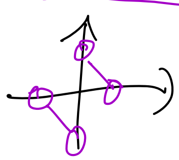
9. **R/A** Complete the following sentences.

(generic)

(a) If $f(x)$ is defined on $(0, L)$, then the Fourier sine series of f is the Fourier series of

$\boxed{\text{odd extension of } f \text{ to } (-L, L)}$



for Fourier sine:

$$b_n = \frac{1}{L}\int_{-L}^{L} \underbrace{\overbrace{f_o(x)}^{\text{odd}}\ \overbrace{\sin\left(\frac{n\pi x}{L}\right)}^{\text{odd}}}_{\text{even}}\,dx$$

$x^3 \cdot x^5 = x^8$

odd · odd = even

$$= 2\cdot\frac{1}{L}\int_0^L f_o(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

if $0<x<L$, $f_o(x) = f(x)$

$$= \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

(b) If $f(x)$ is defined on $(0, L)$, then the Fourier cosine series of f is the Fourier series of

$\boxed{\text{even extension of } f}$

Suppose now g is a function defined on $(-L, L)$, and f is the same function, but considered on $(0, L)$.

(c) If g is an even function on $(-L, L)$, then the ___even___ extension of f to $(-L, L)$ equals g, so the Fourier ___cosine___ series of f on $[0, L]$ is the Fourier series of g on $[-L, L]$.

(d) If g is an odd function on $(-L, L)$, then the ___odd___ extension of f to $(-L, L)$ equals g, so the Fourier ___sine___ series of f on $[0, L]$ is the Fourier series of g on $[-L, L]$.

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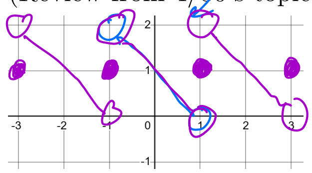
(10.3-10.4) What function does the Fourier series, Fourier sine series, or Fourier cosine series of a function converge to

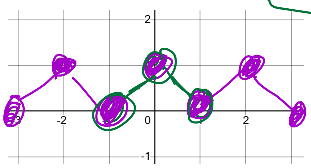
10. **A** As in Exercise 9, let $f(x)$ be the function $1-x$ with domain $(-1,1)$. Let $g(x)$ be the function $1-x$ with domain $(0,1)$. Find formulas for and graphs of each of the following functions for $-3 \le x \le 3$.

    f in blue

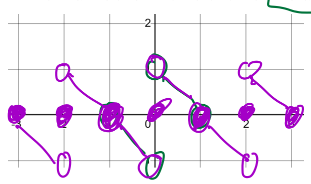
    (a) (Review from 4/30's topic review) The function that the Fourier series of f converges to.

    (b) The function that the Fourier cosine series of g converges to. ← Fourier series of even extension of ~~f~~ g

    (c) The function that the Fourier sine series of g converges to. Fourier series of odd extension of ~~f~~ g

11. **A** Write brief summaries for how to do each of the following:

    (a) (Review from 4/30's topic review) If f is a piecewise-continuously differentiable function on $[-L, L]$, graphically find the function that the Fourier series of f converges to.

    ① Graph f from −L to L.
    ② Copy its graph to make it 2L-periodic
    ③ At each point of discontinuity, the value is the midpoint between left & right limits

    (b) If f is a piecewise-continuously differentiable function on $(0, L)$, graphically find the function that the Fourier cosine series of f converges to.

    ① Graph f from 0 to L
    ② Flip over y-axis to get even extension of f
    ③ Copy its graph to make it 2L-periodic
    ④ At each point of discontinuity, the value is the midpoint between left & right limits

    (c) If f is a piecewise-continuously differentiable function on $(0, L)$, graphically find the function that the Fourier sine series of f converges to.

    Same as (b), but with odd extension instead of even.

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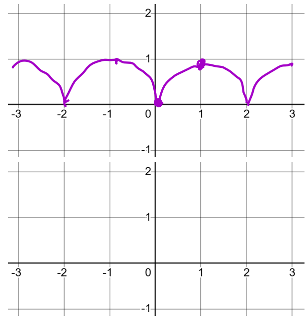
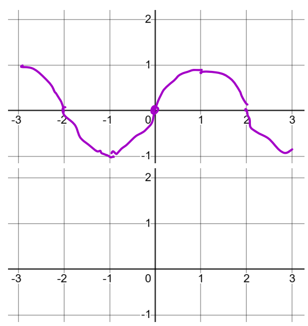
12. **A** Extra practice: Let $f(x) = \sqrt{x}$ for $0 \le x \le 1$. Find graphs of each of the following functions for $-3 \le x \le 3$.

   (a) The function that the Fourier cosine series of f converges to.

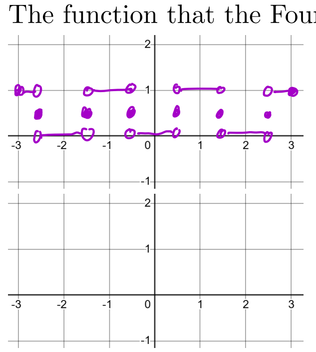


   (b) The function that the Fourier sine series of f converges to.

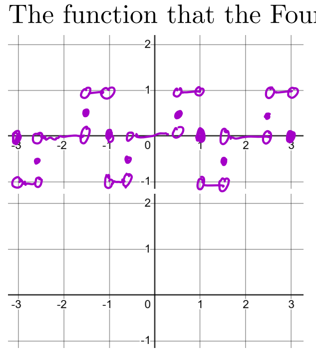


13. **A** Extra practice: Let $f(x) = \begin{cases} 0 & \text{if } 0 < x < \frac{1}{2}, \\ 1 & \text{if } \frac{1}{2} \le x < 1. \end{cases}$. Find graphs of each of the following functions for $-3 \le x \le 3$.

   (a) The function that the Fourier cosine series of f converges to.



   (b) The function that the Fourier sine series of f converges to.



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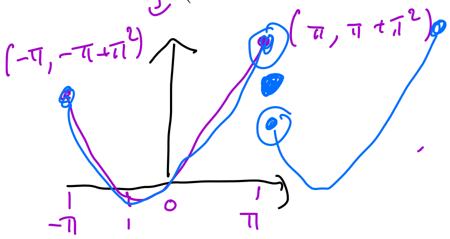
14. **A** Let $f(x) = x + x^2$ on $[-\pi, \pi]$. It turns out that $f \sim \frac{\pi^2}{3} + \sum_{n=1}^{\infty}\left[\frac{4}{n^2}(-1)^n\cos(nx) - \frac{2}{n}(-1)^n\sin(nx)\right]$.

graph of f.

(You should check that if you need the practice.) By plugging in an appropriate value for x, evaluate $\sum_{n=1}^{\infty}\frac{1}{n^2}$.

what would help with this?



$x = 0$?

$\frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2}(-1)^n \cancel{\cos(0)} - \frac{2}{n}(-1)^n \sin 0$

$\frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{4}{n^2}(-1)^n$ $\quad -\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \dots$

Plug in $x = \pi$ into Fourier series: $f \sim \frac{\pi^2}{3} + \sum_{n=1}^{\infty}\left[\frac{4}{n^2}\cancel{(-1)^n}(-1)^n - \frac{2}{n}(-1)^n \sin(n\pi)\right]$

$\sim \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{1}{n^2}$. Also $\frac{1}{2}\left(f(\pi) + f(-\pi)\right) = \frac{1}{2}\left(\pi + \pi^2 + -\pi + \pi^2\right) = \pi^2$.

$\pi^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{1}{n^2}$

$\boxed{\sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}}$

15. Extra practice:

(a) **R** Use the Fourier cosine series of $f(x) = x^2$ on $[0, \pi]$ to show that $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$ equals $\frac{\pi^2}{12}$.

(b) **A** Use the Fourier series of $f(x) = \begin{cases} 0 & -\pi \le x < 0 \\ \pi & 0 \le x < \pi \end{cases}$ on $-\pi < x < \pi$ to show that $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{2n-1} = \frac{\pi}{4}$.

(a) Get FCS of f from Exercise 3e. Plug in x=0 & rearrange.

(b) $f \sim \frac{\pi}{2} + \sum_{n=1}^{\infty}\frac{1}{n}\left(1 - (-1)^n\right)\sin(nx) = \frac{\pi}{2} + \frac{2}{1}\sin(x) + \frac{2}{3}\sin(3x) + \frac{2}{5}\sin(5x) + \dots$

Plug in $x = \frac{\pi}{2}$.

How to identify what would most help you to study next:

* what you don't already know
* & is important.

When looking at a list of "important topics", don't just look at names! Think about what is involved in doing them / see what gets re-used in multiple places.

→ See examples on next page;

Most Important Topics: # for Prof. Sharma's lecture

Modify as needed for Prof. Serganova. (Most should be the same; See what she spent extra time on in lecture OR what she assigned multiple problems.)

From https://edstem.org/us/courses/90195/discussion/8010503

1. Row Echelon/Reduced Row Echelon Form/Elementary Row Operations - **1.1/1.2**

2. Solving Linear Systems - **1.1**

3. Linear/Matrix Transformations - **1.8/1.9**

4. Linear Independence/Dependence/Bases - **1.7/4.3**

5. Inverse Matrix - **2.2**

6. Determinants - **3.1/3.2**

7. Vector Spaces - **4.1**

8. Dimension + Rank/Nullity - **4.4/4.5**

9. Eigenvalues/Characteristic Polynomial - **5.1/5.2**

10. Regular Diagonalization - **5.3**

Find Eigenvector

11. Orthogonality - **6.1/6.2/6.3**

12. Gram-Schmidt Process - **6.4**

Gram-Schmidt

13. Singular Value Decomposition - **7.4**

14. Solving Second Order Homogeneous Linear Equations with Constant Coefficients (Characteristic Equation) - **B4.4**

15. Solving Second Order Nonhomogeneous Linear Equations with Constant Coefficients (Undetermined Coefficients) - **B4.5**

16. Solving Systems of Linear Differential Equations - **B9.5/B9.6**

Medium Importance Topics:

1. Pivots - **1.2**

2. One-to-one/onto transformations/functions - **1.9**

3. Subspaces - **4.1**

4. Null Space/Column Space/Row Space - **4.2**

5. Similar Matrices - **5.2**

6. Pseudo-Diagonalization of 2×2 real matrices with nonreal eigenvalues - **5.5**

7. Orthogonal Projections - **6.3**

8. QR Factorization - **6.4**

9. Least Squares Curves - **6.5/6.6**

10. Inner Product Spaces - **6.7**

11. Symmetric Matrices/Diagonalization of Symmetric Matrices - **7.1**

12. Superposition Principle - **B4.5**

13. Variation of Parameters - **B4.6**

Prof. Sharma only

14. Mixing Problems - **B9.5**

Prof. Serganova did Wronskians in Ch. 9 instead

15. Fourier Series (generic, sin, and cos) - **B10.3/B10.4**

Doesn't mean "ignore"
→ Prof. Sharma still asked something about 4.4 as Q1 on MT2,
→ Prof. Serganova may not fully agree with this list.

Low Importance Topics:

1. Transformations on R^2 (shear, dilation, etc.) - **1.9**

2. Cramer's Rule - **3.3**

3. Adjugate Matrix - **3.3**

4. Physical applications of determinant (volume etc.) - **3.3**

5. Coordinate Systems - **4.4**

6. Change of Basis - **4.6**

7. Quadratic Forms - **7.2**

8. High order differential equations as system of first order equations - **B9.1**

9. Addendum Section

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Finding eigenvectors of 3x3 matrices from their eigenvalues

Green is computational ideas, purple is what I'd write!

16. **R** The matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ -4 & -2 & 1 \\ -1 & -1 & -1 \end{bmatrix}$ has eigenvalues -1 and $-1 \pm 2i$. Find an eigenvector for each eigenvalue.

$$A-(-1+2i)I = \begin{bmatrix} 1-2i & 1 & 0 \\ -4 & -1-2i & 1 \\ -1 & -1 & -2i \end{bmatrix} \xrightarrow{R_2 \to R_2 - 4R_3} \begin{bmatrix} \cancel{1-2i} & \cancel{1} & \cancel{0} \\ 0 & 3-2i & 1+8i \\ -1 & -1 & -2i \end{bmatrix}$$

I can cross out this row because, if $-1+2i$ is an eigenvalue of A, then these 3 rows must be linearly dependent. The bottom two rows are linearly independent. (If you switched them, bottom two rows would be in echelon form) So the top row must be a linear combination of the bottom two rows, so if we kept on going (by switching rows 1 & 3), eventually, the 3rd row would be all 0s.

Row 2: pick $x_3 = 3-2i$, so $x_2 = -1-8i$

Reason: Row 2 says:

$$(3-2i)x_2 + (1+8i)x_3 = 0$$

So $x_2 = -\frac{1+8i}{3-2i}x_3$,

Pick $x_3 = 3-2i$ to cancel the denominator.

Row 3: $-x_1 - x_2 - 2ix_3 = 0$

$$\begin{aligned} x_1 &= -x_2 - 2ix_3 \\ &= 1+8i-2i(3-2i) \\ &= 1+8i-6i-4 \\ &= -3+2i. \end{aligned}$$

$$\boxed{\text{Answer: for } \lambda=-1+2i:\ \vec{x} = \begin{bmatrix} -3+2i \\ -1-8i \\ 3-2i \end{bmatrix}}$$

You try the rest.

Check: (Because crossing out a row means you lose a way to check if your eigenvalue is correct).

$$\begin{bmatrix} 0 & 1 & 0 \\ -4 & -2 & 1 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} -3+2i \\ -1-8i \\ 3-2i \end{bmatrix} = \begin{bmatrix} -1-8i \\ (12-8i)+(2+16i)+(3-2i) \\ (3-2i)+(1+8i)+(-3+2i) \end{bmatrix} = \begin{bmatrix} -1-8i \\ 17+6i \\ 1+8i \end{bmatrix}$$

Extra practice: Repeat: $\begin{bmatrix} -2 & 1 & -1 \\ -2 & -2 & 0 \\ -1 & 0 & -2 \end{bmatrix}$ $(-2, -2\pm i)$; $\begin{bmatrix} 5 & -1 & -3 \\ -5 & 4 & -5 \\ -5 & 2 & -3 \end{bmatrix}$ $(2, 2\pm i)$; $\begin{bmatrix} -7 & -15 & 15 \\ -3 & -1 & 9 \\ -4 & -7 & 10 \end{bmatrix}$ $(-2, 2\pm 3i)$.

For even more practice, repeat with the transposes of the four matrices on this page (A and A^T will have the same eigenvalues, but not always the same eigenvectors... why?)

$$(-1+2i)\begin{bmatrix} -3+2i \\ -1-8i \\ 3-2i \end{bmatrix} = \begin{bmatrix} 3-2i-6i-4 \\ 1+8i-2i+16 \\ -3+2i+6i+4 \end{bmatrix} = \begin{bmatrix} -1-8i \\ 17+6i \\ 1+8i \end{bmatrix}$$

Same, so $A\vec{x} = (-1+2i)\vec{x}$, so $\vec{x}$ is actually an eigenvector.

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Gram-Schmidt (and inner products)

Key computational steps in green.

17. **R** Find an orthogonal basis for Col A where $A = \begin{bmatrix} -1 & 9 & 2 \\ 3 & -7 & 1 \\ 1 & -1 & 2 \\ 1 & -5 & -1 \end{bmatrix}$.

$$\begin{array}{ccc} \uparrow & \uparrow & \uparrow \\ u_1 & u_2 & u_3 \end{array}$$

$$\boxed{\vec{v}_1 = \vec{u}_1 = \begin{bmatrix} -1 \\ 3 \\ \vdots \end{bmatrix}}$$

$$\vec{u}_2 - \frac{\vec{v}_1 \cdot \vec{u}_2}{\vec{v}_1 \cdot \vec{v}_1}\vec{v}_1 = \vec{u}_2 - \frac{-9-21-1-5}{1+9+1+1}\vec{v}_1$$

$$= \vec{u}_2 - \frac{-36}{12}\vec{v}_1 = \vec{u}_2 + 3\vec{v}_1 = \begin{bmatrix} 9 \\ -7 \\ -1 \\ -5 \end{bmatrix} + \begin{bmatrix} -3 \\ 9 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 2 \\ -2 \end{bmatrix}.$$

Rescale; divide by 2 to get smaller numbers; pick $\boxed{\vec{v}_2 = \begin{bmatrix} 3 \\ \vdots \end{bmatrix}}$

$$\vec{u}_3 - \frac{\vec{v}_1 \cdot \vec{u}_3}{\vec{v}_1 \cdot \vec{v}_1}\vec{v}_1 - \frac{\vec{v}_2 \cdot \vec{u}_3}{\vec{v}_2 \cdot \vec{v}_2}\vec{v}_2 = \vec{u}_3 - \frac{-2+3+2-1}{1+9+1+1}\vec{v}_1 - \frac{6+1+2+1}{9+1+1+1}\vec{v}_2$$

$$= \vec{u}_3 - \frac{2}{12}\vec{v}_1 - \frac{10}{12}\vec{v}_2.$$

Rescale: Multiply by 6 to clear fractions:

$$6\vec{u}_3 - \vec{v}_1 - 5\vec{v}_2 = \begin{bmatrix} 12 \\ 6 \\ 12 \\ -6 \end{bmatrix} + \begin{bmatrix} 1 \\ -3 \\ -1 \\ -1 \end{bmatrix} + \begin{bmatrix} -15 \\ -5 \\ -5 \\ 5 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \\ 6 \\ -2 \end{bmatrix}.$$

Rescale: Divide by 2 to get smaller numbers. Pick $\boxed{\vec{v}_3 = \begin{bmatrix} -1 \\ -1 \\ 3 \\ \vdots \end{bmatrix}}$ $\boxed{\text{Ans: } \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}}$

Extra practice: Repeat with one of matrices in Exercises 6.4.9-6.4.12, but with some of the columns swapped around. Find the orthogonal projection of $\begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix}$ onto Col $\begin{bmatrix} 2 & -1 \\ 1 & 3 \\ 2 & -1 \end{bmatrix}$. Find an orthogonal basis of Span$\{1, x, x^2\}$ under the inner product $< f, g > = \int_0^1 f(x)g(x)$.

You do the rest.